

MailTrace AI - Target Users & Complete Dashboard Guide

Enterprise Threat Detection, Origin Geolocation & Forensic Telemetry Platform

Version 1.0.0 | SIH Production Edition

PART 1: TARGET USERS & ORGANIZATIONAL PERSONAS

1. Tier 1 & Tier 2 SOC Analysts (Security Operations Centers)

- Rapidly triages suspicious employee emails without manual header decoding.
- Deconstructs zero-width obfuscations, masked URLs, and credential lures in seconds.
- Exports court-admissible forensic PDF evidence reports for ticketing systems.

2. Chief Information Security Officers (CISOs) & Security Directors

- Monitors enterprise-wide risk exposure across departments and branch networks.
- Tracks coordinated adversary attack campaigns targeting corporate identities.
- Evaluates corporate security policy compliance and automated containment metrics.

3. Digital Forensics & Cybercrime Investigators (DFIR / Law Enforcement)

- Extracts genuine sending nodes, bypassing internal corporate relay proxies.
- Traces multi-hop chronology and classifies Tor exit nodes / Bulletproof VPS hosting.
- Validates tamper-proof custody chains backed by SHA-256 and Web3 blockchain seals.

4. Corporate Mail & IT Administrators (Exchange / Google Workspace)

- Connects live company mailboxes via IMAP Sentinel for automated background ingestion.
- Inspects live DNS records, MX health, SPF/DKIM keys, and DMARC enforcement.

PART 2: UNIFIED DASHBOARDS & MODULE GUIDE (Page 1 of 2)

Dashboard 1: Scan Email & Verification Lab (/ or /verify)

- Primary manual ingestion gateway supporting .eml, .msg files and raw MIME paste.
- 5-Stage Animated Parsing HUD: MIME validation -> Crypto Auth -> URL Deobfuscation -> Homoglyph inspection -> Attack DNA correlation.

Dashboard 2: Security SOC Dashboard (/dashboard)

- Executive telemetry hub featuring 9 real-time metrics, geographical sending nodes map, live threat feed, and 3-pathway ingress onboarding banner when in standby.

PART 2: UNIFIED DASHBOARDS & MODULE GUIDE (Continued)

Dashboard 3: Live Mailbox Sentinel (/live)

- Continuous zero-trust background mailbox interception for Gmail, Outlook, Yahoo, and IMAP.
- Background polling every 15s with 1-click Disconnect, on-demand Sync, and DNS Tool.

Dashboard 4: Deep Threat Verdict & Explainable Score (/result/:id)

- Comprehensive 0-100 risk dial with mathematical penalty breakdowns.
- Natural Language Social Engineering meters: Urgency, Credential Harvesting, Financial Fraud.
- Interactive Investigation Tabs: Attack Story Timeline, Infrastructure Graph, Link Security, Origin GeoTrace, Attack DNA, and Printable PDF Forensic Report Export.

Dashboard 5: Email Forensics Lab (/forensics)

- Multi-Hop Route Timeline showing hop latencies and intermediate mail transfer agents (MTAs).
- Cryptographic Authentication Matrix (SPF / DKIM / DMARC) and Raw RFC-5322 Viewer.

Dashboard 6: Origin GeoTrace & GPS Intelligence (/geotrace)

- Pinpoints earliest reliable sender IP with Leaflet & Esri Dark Canvas GPS map.
- Classifies Autonomous System (ASN), ISP, and flags Proxy / Tor Exit / Hosting Relays.

Dashboard 7: Attack DNA & Campaign Clusters (/attack-dna)

- Groups isolated phishing emails sharing origin infrastructure into adversary clusters.
- Dynamic Evolution Wave Tracker analyzing attacker adaptations across multi-wave campaigns.

Dashboard 8: Security Policies & Case Incidents (/incidents)

- Automated SOC triage case ticketing when emails violate security rules or risk thresholds.
- Interactive workflow status management: OPEN -> INVESTIGATING -> CONTAINED -> CLOSED.

PART 2: UNIFIED DASHBOARDS & MODULE GUIDE (Final)

Dashboard 9: Web3 Blockchain Evidence Ledger (/ledger)

- Anchors SHA-256 evidence digests onto the Polygon Amoy EVM Distributed Ledger.
- Independent Verifier Tool: Input any hash or drop an .eml file to verify on-chain seal.
- Generates cryptographic notarization certificates for legal non-repudiation.

Dashboard 10: Audit History Archive (/history)

- Complete searchable database of all scanned emails with instant search, threat pills, and 1-click deletion with automatic campaign re-clustering.

Dashboard 11: ARIA AI Security Assistant (Floating Copilot)

- Context-aware conversational AI assistant powered by Google Gemini.
- Navigates pages, explains email authentication tokens, and triages threat vectors.

SUMMARY OF ARCHITECTURAL CAPABILITIES

Feature	Implementation Status	Protocol / Technology

RFC Header Parsing	Production Active	RFC-5322 MIME Parser
Cryptographic Auth Matrix	Production Active	SPF, DKIM, DMARC Validator
Origin Geolocation	Production Active	Earliest IP + Leaflet GPS
Social Engineering NLP	Production Active	Urgency & BEC Heuristic Engine
Attack DNA Correlation	Production Active	Jaccard Cluster Clustering
Live Mailbox Ingestion	Production Active	IMAP4 SSL Background Daemon
Web3 Notarization	Production Active	Polygon Amoy EVM Registry
Cloud Database Sync	Production Active	Supabase PostgreSQL + RLS
AI SOC Assistant	Production Active	Google Gemini 1.5 Flash API

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